Support Material-KDD-1060

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1 Additional Experiment Results to Reviewer gq6s

W1/W5: Real-data faithfulness and representative patches

Table 1 summarizes the real-data faithfulness results. It reports overall statistics and dataset-level `I(20)`/`I(100)` metrics under a fixed masking-and-retraining protocol. These results evaluate temporal faithfulness beyond synthetic localization. Under the stringent `I(20)` threshold, AnchorMoE achieves **0.6419**. This exceeds both the strongest TS-specific baseline (FIT, **0.5760**) and the generic baseline (IG, **0.6189**). Under `I(100)`, AnchorMoE reaches **0.8620**, and its gradient-free variant reaches **0.8667**, both outperforming Dynamask (**0.8284**). These findings demonstrate strong temporal faithfulness on real data, though we acknowledge they do not inherently prove full semantic interpretability.

Figure 1 provides a representative NATOPS case study to complement Table 1. It visually illustrates the quantitative findings: the highest-ranked temporal regions are concentrated within a small, specific subset of the sequence rather than scattered diffusely.

2 Additional Experiment Results to Reviewer Hjgq

W3: Gradient-role control under real-data faithfulness

Table 2 evaluates the impact of UADB's gradient reliance. Under the real-data faithfulness protocol, the gradient-free variant of AnchorMoE still outperforms IG on mean `I(100)` (**0.8667** vs. **0.8384**). Furthermore, the standard AnchorMoE model remains superior under the restrictive `I(20)` threshold (**0.6419** vs. **0.6189**). This confirms that gradient magnitude serves as a training-time calibration cue, not as the final inference explanation. (Refer to Table 1 for complete dataset-level statistics).

Table 2: Gradient-role control analysis. Higher is better for `I(20)` and `I(100)`; lower is better for rank.

Method	I(20)	Rank	I(100)	Rank
AnchorMoE	**0.6419**	**2.57**	0.8620	**2.29**
AnchorMoE-gradient-free	0.6338	3.14	**0.8667**	2.43
IG	0.6189	2.71	0.8384	3.14

W2/W4/W5/W7/W8/W9: Mechanism diagnostics and internal design choices

Tables 3–6 provide numerical details for the mechanism diagnostics. Table 3 reports aggregate statistics across 29 datasets. Tables 4 and 5 highlight representative per-dataset utilization and κ distributions, respectively. Table 6 reports the B2 routing diagnostics, where removing DPAR increases inter-anchor redundancy from **0.3488** to **0.7213**.

Table 7 reports targeted B1/B2/B3 internal ablations under a diagnostic setup; within B1, removing the sample-level relevance cue lowers mean Acc from **0.9686** to **0.7084**. Table 11 reports the binding-specific diagnostics: NoConfidence yields an all-one κ regime and lowers mean `I(20)` from **0.8953** to **0.8819**. Together with Table 6, these diagnostics provide empirical support for the query design, redundancy control, and confidence-weighted binding.

Detailed sensitivity analysis (W9) is provided in Section 4.

3 Additional Experiment Results to Reviewer gMhY

W1: Scalability and explanation-aware efficiency

Table 8 and Figure 3 report the 10k-scale points under explanation-aware accounting. Specifically, AnchorMoE achieves **122.8** iter/s for the 10k feature-dimension point (vs. **30.4** for DLinear), and **57.1** iter/s for the 10k sequence-length point (vs. **6.7** for TSLANet). Crucially, the reported throughput includes the computational cost of post-hoc explanations for the black-box baselines, ensuring a fair operational comparison between ante-hoc and post-hoc systems.

Table 8: Explanation-aware throughput and peak memory for 10k-scale points. Throughput is measured in inference-plus-explanation iter/s (higher is better); memory in MB (lower is better).

Sweep	Point	AnchorMoE	Peak MB	Best baseline	Baseline	Rank
feature_dim	10000	122.822	535.1	DLinear	30.430	1
seq_len	10000	57.059	230.3	TSLANet	6.659	1

W2: Temporal faithfulness on real data

We apply the same real-data faithfulness protocol discussed in Section 1. Please refer to Table 1 and Figure 1 for evidence supporting temporal faithfulness on real-world datasets.

4 Additional Experiment Results to Reviewer ncgk

W1/W3: Hyperparameter stability and anchor-count selection

Table 9, Figure 4, and Table 10 address hyperparameter stability. In the reported sweeps, `anchor_div`, `spectral_weight`, and `num_groups` exhibit broad workable regions, whereas `conf_lambda` requires conservative tuning. For instance, on the complex NATOPS dataset, aggressive confidence regularization causes accuracy to drop sharply from **0.9500** to **0.1667**. We therefore recommend a plateau-based heuristic for anchor-count selection: choose the smallest group count that reaches the validation plateau. As shown in Table 10, the optimal `num_groups` consistently falls within a moderate range (**3**, **4**, or **5**).

Table 1: Real-data faithfulness evaluation. The top block reports overall means and mean ranks; the lower block details dataset-level metrics. Higher is better for `I(20)` and `I(100)`; lower is better for rank.

Dataset / Stat.	Metric	AnchorMoE	AnchorMoE-gradient-free	IG	DeepLift	Grad-CAM	Random	Dynamask	FIT
Summary statistics									
Overall mean	I(20)	**0.6419**	0.6338	0.6189	0.6168	0.5891	0.5301	0.5390	0.5760
Overall rank	I(20)	**2.86**	3.43	3.14	3.86	4.71	6.29	6.00	5.71
Overall mean	I(100)	0.8620	**0.8667**	0.8384	0.8363	0.8278	0.8035	0.8284	0.8264
Overall rank	I(100)	**2.57**	2.86	3.71	4.43	5.86	6.29	4.57	5.71
Dataset-level values									
ArticularyWordRecognition	I(20)	0.6554	0.6625	**0.6808**	0.6749	0.5832	0.6749	0.5757	0.6361
ArticularyWordRecognition	I(100)	0.8772	0.8810	0.8812	**0.8963**	0.8599	0.8938	0.8812	0.8725
BasicMotions	I(20)	0.5750	**0.5938**	0.4625	0.5094	0.5521	0.4583	0.5083	0.4531
BasicMotions	I(100)	0.8175	**0.8675**	0.7354	0.7306	0.7792	0.7388	0.8092	0.7819
Cricket	I(20)	0.6597	0.6111	0.6985	**0.6997**	0.6140	0.6198	0.6024	0.6869
Cricket	I(100)	0.8993	0.8819	0.9041	**0.9078**	0.8703	0.8443	0.8760	0.8892
FordA	I(20)	0.6661	0.6821	0.6301	0.6412	0.6469	0.4620	**0.6920**	0.5631
FordA	I(100)	0.8678	0.8766	0.8401	0.8287	0.8541	0.7505	**0.8800**	0.8061
NATOPS	I(20)	0.6764	0.6708	**0.6792**	0.6769	0.6069	0.5315	0.4843	0.5486
NATOPS	I(100)	**0.8883**	0.8808	0.8744	0.8671	0.8517	0.8500	0.7591	0.8252
RacketSports	I(20)	**0.5576**	0.5206	0.4729	0.4482	0.5414	0.4238	0.4833	0.4512
RacketSports	I(100)	**0.7566**	0.7541	0.7054	0.7053	0.7194	0.6762	0.7278	0.6910
SpokenArabicDigits	I(20)	0.7030	0.6959	**0.7081**	0.6675	0.5790	0.5405	0.4271	0.6928
SpokenArabicDigits	I(100)	0.9274	0.9251	**0.9283**	0.9180	0.8600	0.8711	0.8658	0.9191

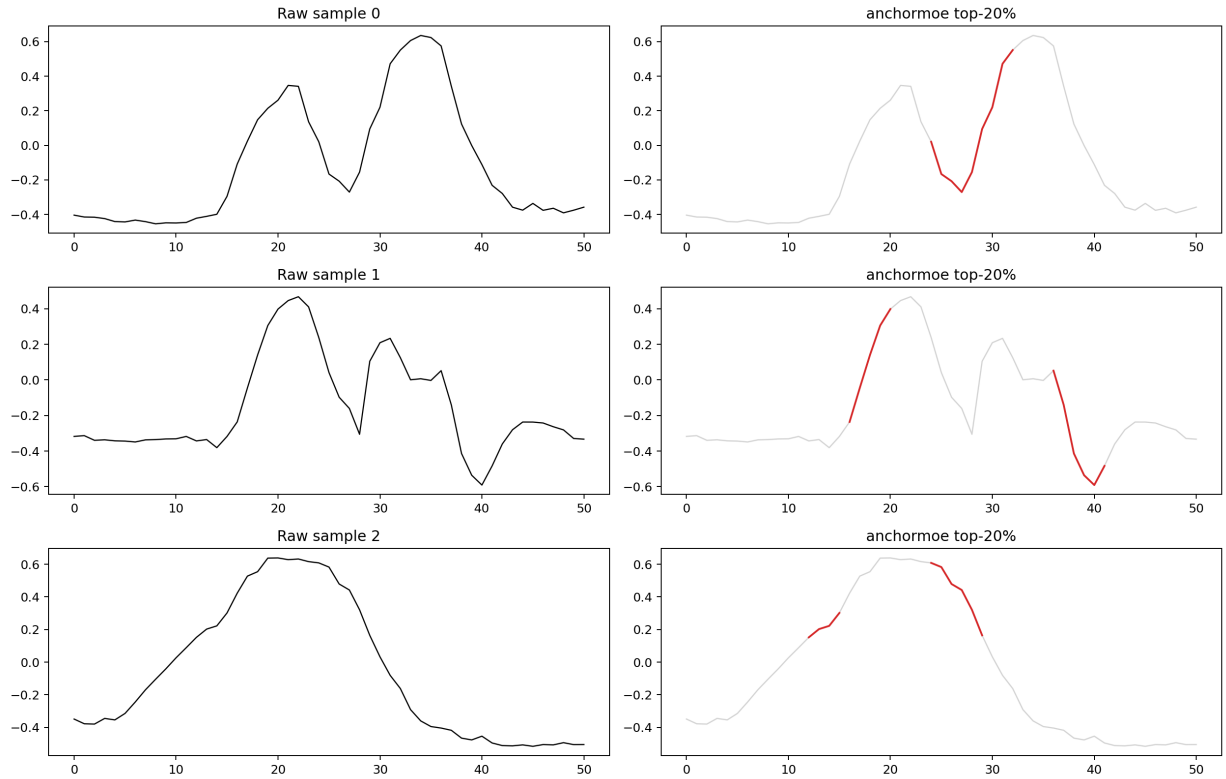


Figure 1: Representative NATOPS case study. Temporal evidence highlighted by AnchorMoE is compact and localized.

Table 3: Aggregate mechanism statistics across the 29-dataset diagnostic sweep.

Metric	Mean	Median	Min	Max
active_expert_rate	0.5388	0.5000	0.1667	1.0000
max_expert_share	0.7674	0.7578	0.4705	1.0000
effective_experts	1.8952	1.9053	1.0000	3.2933
κ_mean	0.1844	0.1662	0.0001	0.6128
κ_low_ratio	0.5653	0.6325	0.0000	1.0000
κ_high_ratio	0.0272	0.0002	0.0000	0.2603

Table 4: Per-dataset expert utilization. The model adapts to both distributed and concentrated routing regimes depending on the dataset.

Dataset	Active rate	Max share	Effective experts
Libras	1.0000	0.4932	3.2933
Cricket	0.6667	0.5594	3.0690
PhonemeSpectra	0.3750	0.4705	2.9461
BasicMotions	0.7500	0.5656	2.2905
Heartbeat	0.2500	1.0000	1.0003
MotorImagery	0.1667	1.0000	1.0000

Table 5: Per-dataset κ statistics. The gating mechanism adapts to varying sparsity regimes across different datasets.

Dataset	κ mean	Low ratio	High ratio
Libras	0.61276	0.07514	0.22591
Handwriting	0.44180	0.33335	0.26026
NATOPS	0.11251	0.81994	0.03575
SpokenArabicDigits	0.08893	0.86469	0.03999
Heartbeat	0.01114	0.97854	0.00000
MotorImagery	0.00007	0.99996	0.00000

Table 6: B2 routing diagnostics. Removing DPAR causes a severe 106.8% increase in inter-anchor redundancy.

Variant	Active rate	Max share	Effective experts	κ mean	Low ratio	High ratio	Redundancy
Full	0.4389	0.8525	1.4280	0.5319	0.1483	0.2606	0.3488
NoDPAR	0.5470	0.8283	1.4563	0.5600	0.1432	0.2865	0.7213
WeakDPAR	0.4278	0.8432	1.4518	0.5306	0.1528	0.2714	0.3788

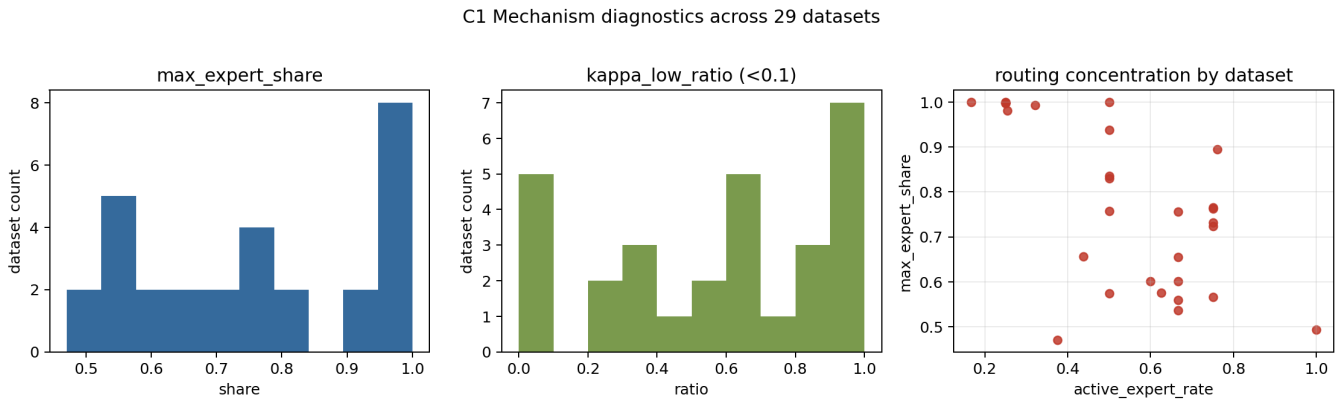
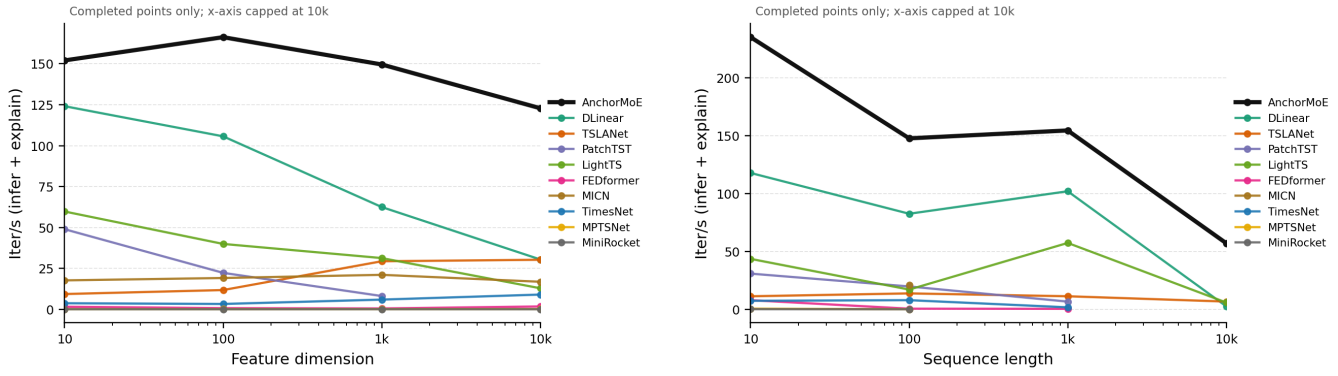


Figure 2: Mechanism diagnostics across 29 datasets, highlighting dataset-dependent routing concentration and non-trivial κ behavior.



(a) Feature-dimension sweep. **(b) Sequence-length sweep.**

Figure 3: Explanation-aware throughput trends evaluated up to the 10k scale.

Table 7: Targeted internal-design ablations probing the proposed query, routing, and binding choices under a diagnostic setup. These block-wise results are not directly comparable to the 18-dataset overall averages reported in Table 2 of the main paper. Higher is better for all metrics.

Block	Variant	Mean Acc	Mean F1	Mean I(20)	Mean I(100)
B1	Full	0.9686	0.9685	0.9250	0.9348
B1	NoRelevance	0.7084	0.6687	0.8478	0.9348
B1	NoSpectral	0.9737	0.9736	0.9268	0.9348
B2	Full	0.9686	0.9685	0.9250	0.9348
B2	NoDPAR	0.9727	0.9726	0.9580	0.9348
B2	WeakDPAR	0.9734	0.9732	0.9518	0.9348
B3	Full	0.9509	0.9504	0.8953	0.9081
B3	AnchorMoE-gradient-free	0.9638	0.9638	0.9097	0.9051
B3	NoConfidence	0.9477	0.9475	0.8819	0.9078

Table 9: Sensitivity analysis summary. In the reported sweeps, `anchor_div`, `spectral_weight`, and `num_groups` show broader workable ranges, whereas `conf_lambda` is the most sensitive factor.

Parameter	Scanned values	Acc. range	Reading
anchor div.	0, 0.02, 0.05, 0.08, 0.1	0.9268–0.9332	Broadly workable in the current sweep.
spectral weight	0, 0.1, 0.3, 0.5, 0.7	0.9281–0.9374	Moderate values remain stable in aggregate.
num groups	2, 3, 4, 5, 6, 7, 8	0.9166–0.9722	Use the smallest count already on the plateau.
conf lambda	0, 0.005, 0.01, 0.02, 0.03	0.7976–0.9314	Most sensitive factor; NATOPS drops to 0.1667 under aggressive penalization.

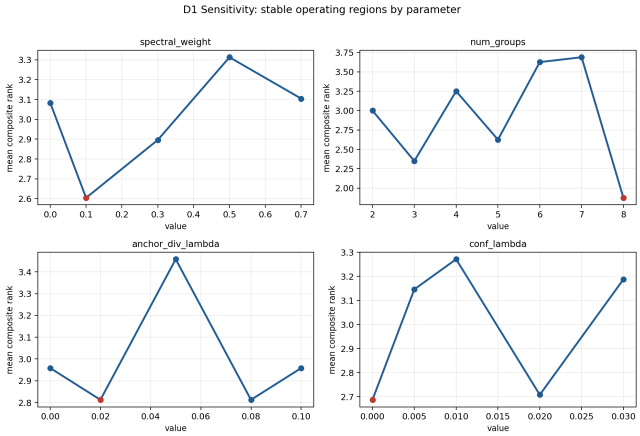


Figure 4: Stable operating regions for the primary structural hyperparameters.

W2: Selective binding and deployment caveat

Refer to Table 1 for real-data faithfulness statistics. Table 11 highlights binding-specific diagnostics: ablating confidence weighting (NoConfidence) forces an all-one gating regime and lowers mean `I(20)` from **0.8953** to **0.8819**. This supports the role of a non-trivial confidence-weighted binding signal.

Table 10: Dataset-specific optimal `num_groups`. The optimum stays within a moderate range (3–5).

Dataset	Best num_groups	Acc	F1	AOPCAcc	IoU
ArticularyWord Recognition	3	0.9733	0.9733	0.7898	0.2820
Cricket	5	0.9722	0.9720	0.7000	0.2796
FordA	3	0.9356	0.9356	0.3461	0.2499
Heartbeat	4	0.7707	0.7674	0.4898	0.2769
NATOPS	4	0.9833	0.9833	0.6578	0.3750
SpokenArabic Digits	3	0.9900	0.9900	0.7773	0.3007

Table 11: Selective-binding diagnostics. NoConfidence yields an all-one gate and lower early-stage faithfulness.

Variant	I(20)	I(100)	κ	Low	High
Full	0.8953	0.9081	0.3833	0.0989	0.0504
AnchorMoE-gradient-free	0.9097	0.9051	0.2322	0.4188	0.0049
NoConfidence	0.8819	0.9078	1.0000	0.0000	1.0000